

HARSHA KEERTHAN MUTHU KRISHNAN

+65 81134132 · m.harshakeerthan@gmail.com · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

EXPERIENCE

NCS Software Engineer, Engineering

Sep 2024 - Present

- AI-Enabled Aviation Dashboard (NCS Impact 2025 Showcase):
 - Engineered a containerised geospatial dashboard (MapLibre, PostGIS, self-hosted tiles) integrating 5+ aviation data sources into a unified real-time map for ops scenarios.
 - Selected as one of the showcases for the enterprise pillar to demonstrate the platform's capabilities to senior government leadership (incl. Deputy Prime Minister, ~1,600 attendees) following successful demos to CAAS.
- Foundational Java & Enterprise Application Support:
 - Completed Nucleus Bootcamp Program (Java Track) and delivered a capstone project using Java, Maven, Spring Boot, REST APIs, JPA/Hibernate, RBAC, SQL.
 - Resolved over 10 production bugs and change requests on a critical Alarm Management system for the Land Transport Authority (LTA) and added 15 JUnit tests, structured logging to prevent repeats.
- Delivered SAML-authenticated LMS with bulk ingest, assessments, certificates and progress tracking. Removed manual excel checks (~6.5 h/mo, est. saved) and migrated 5 courses.
- Built a Network Control Panel to automate infrastructure management, leveraging Ansible playbooks to orchestrate OS patching for Cisco/Fortigate/Linux devices and manage configuration with version history. Demoted to CAG.
- Developed StreamASR, a real-time transcription service in Python to stream audio to an external ASR service. Deployed on AWS EC2 with DynamoDB for session/transcript state.

MediaTek Full Stack Developer Intern

May 2023 - Dec 2023

- Web development: Redesigned intranet websites using the MERN stack and CMS, enhancing user interface and security for over 50 chip designers, improving request turnaround from 7 to 2 days.
- DevOps: Established a local version control system and automated CI/CD pipeline, reducing deployment time by 75% and enhancing collaboration. Implemented automated SSL renewals, ensuring uninterrupted service during certificate updates.
- Scripting: Developed Python scripts to utilize RESTful APIs, automating content synchronization across Drupal websites, ensuring zero redundancy and improving data update efficiency.
- Tech Stack: Python, SQL, PHP, Drupal, React, Express & Node.js, RESTful & JSON API, Agile practices

Continental Automotive Software Engineer Intern

May 2022 - Aug 2022

- Engineered communication between firmware and sensors by implementing UART interface to reduce failure rates to near zero using Python in Linux operating systems.
- Demonstrated to Continental's Business Unit, leading to successful production deployment and market launch.
- Used 3D motion technology to capture machine learning data and stored it in MongoDB using MQTT for real-time analysis.
- Tech Stack: Python, C++, MQTT, CMake, ECal, Linux

PROJECTS

SITAtlas | Final Year Project

- Engineered a **web-based geospatial navigation system** using Mapbox GL JS to address wayfinding challenges in SIT's new Punggol campus, developed with **Next.js, React, and PayloadCMS for a full-stack solution**.
- Integrated an **AI-powered chatbot utilizing OpenAI's GPT-4o model** for user interaction with geospatial data.
- Implemented a robust **CI/CD pipeline** with GitHub Actions, automating **Docker** deployments to Digital Ocean VPS.
- **Published and presented a Conference Paper** at the *2024 IEEE International Smart Cities Conference (ISC2-2024)*.
- Tech Stack: NextJS, TypeScript, Node.js, MongoDB, Mapbox, PayloadCMS, GraphQL, NGINX, Docker, DigitalOcean

FlightLogger

- Developed a flight log management system with **NextAuth for OAuth and JWT-based authentication**, implementing **role-based access control**, and leveraging **Next.js server actions** for efficient data handling.
- Tech Stack: Next.js 14, React, MongoDB, NextAuth.js, Tailwind CSS, Shadcn UI, TypeScript, Resend, Vercel

EDUCATION

Singapore Institute of Technology

Sept 2021 - Sept 2024

B.Eng with Honours in Computer Engineering

Specialisation in Intelligent Transportation Systems (Telematics)

GPA: 4.57 | First Class Honours

Continental Automotive Singapore Book Prize (2024) | SINDA Excellence Award (2025)

SKILLS

Python, C++, Java, Docker, CI/CD, Ansible, React, TypeScript/Node.js, SQL, AWS, Jest, Prometheus/Grafana, OpenAPI